the output to public_html and public_html/admin; and add a small .htaccess to each directory that rewrites unknown paths back to index.html so the Angular router, rather than the web server, decides what a deep link means.

The full procedure, including the exact commands and the problems below, is written up in docs/deployment.md so it can be repeated from scratch.

10. Problems Encountered and What They Taught Us

Four things went wrong during deployment. They are recorded here because each one turned out to be a lesson about the layer beneath the one we thought we were working on.

Every emoji became a question mark. The category and organisation icons (a runner, a bowl, a music note, a house) appeared correctly on localhost and as ? on the server. Our first guess was the SQL import, so we added SET NAMES utf8mb4; it did not help. Our second guess was the driver, so we pinned charset: 'utf8mb4' in the mysql2 pool; necessary, but still not enough. The real cause was one level lower: the CREATE TABLE statements never declared a character set, so the tables inherited the database default, and the host's default was utf8mb3 - three bytes per character, which cannot hold a four-byte emoji at all. It had been failing silently since the tables were created, which is why no error was ever raised. The proof was the medical symbol on the wildlife shelter: it sits inside the three-byte range and it was also corrupted, which ruled out 'emoji' as the category of the bug and pointed at the whole character set. The fix was to convert the tables to utf8mb4 and rewrite the affected values; the lasting fix was to pin the character set in the export so the same thing cannot happen again.

The admin site was blank. It worked locally and served a white page on the server, with no 404 and no error message. An Angular build assumes it is mounted at the root unless it is told otherwise, so index.html asked the browser for /main.js while the file had been uploaded to /admin/main.js. The browser received the HTML page again - the .htaccess rewrite - and tried to parse HTML as JavaScript, which fails without a visible message. Rebuilding with --base-href=/admin/ fixed it. The lesson is that a rewrite rule can hide a broken asset path by answering 200 to everything.